

ORIGINAL RESEARCH

PERIPHERAL

Comparative Efficacy of Endovascular Revascularization Modalities for Symptomatic Infrapopliteal Peripheral Artery Disease



A Network Meta-Analysis

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ABSTRACT

BACKGROUND Patients with infrapopliteal peripheral artery disease (PAD) are at elevated risk for amputation and mortality. The comparative efficacy of available endovascular devices for treating infrapopliteal PAD is unknown.

OBJECTIVES The aim of this study was to compare the efficacy of percutaneous transluminal angioplasty (PTA), bare-metal stents (BMS), drug-coated balloons (DCBs), and drug-eluting stents (DES), including bioresorbable DES, in patients with symptomatic infrapopliteal PAD.

METHODS Online databases were searched for published randomized controlled trials comparing PTA, BMS, DCB, and DES in patients with infrapopliteal PAD. Random-effects models were used to estimate ORs for binary variables and mean differences (MDs) for continuous variables.

RESULTS Twenty-one randomized controlled trials comprising 2,958 patients were included (PTA, n = 1,102; BMS, n = 284; DCBs, n = 903; DES, n = 695). Compared with PTA, both DCBs and DES had higher primary patency (DCB OR: 2.66 [95% CI: 1.52-4.64; $P = 0.001$]; DES OR: 2.41 [95% CI: 1.23-4.73; $P = 0.01$]) and lower target lesion binary (>50%) restenosis (DCB OR: 0.40 [95% CI: 0.21-0.77; $P = 0.006$]; DES OR: 0.28 [95% CI: 0.13-0.62; $P = 0.002$]). DCBs also lowered late lumen loss (MD: 0.34; 95% CI: -0.63 to -0.05; $P = 0.02$), in-lesion diameter stenosis (MD: -10.04; 95% CI: -19.20 to -0.88; $P = 0.03$), and clinically driven target lesion revascularization (OR: 0.47; 95% CI: 0.29-0.76; $P = 0.002$). Conversely, BMS decreased the odds of complete wound healing (OR: 0.41; 95% CI: 0.20-0.85; $P = 0.02$). No significant differences in amputation or mortality were observed across the 4 comparator groups.

CONCLUSIONS Compared with PTA, DCBs and DES demonstrated higher primary patency and lower binary restenosis, and DCBs reduced late lumen loss, percentage in-lesion diameter stenosis, and clinically driven target lesion revascularization. (JACC Cardiovasc Interv. 2026;19:1780-1792) © 2026 by the American College of Cardiology Foundation.

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Lower extremity peripheral artery disease (PAD) affects 6.5 million to 8.5 million people in the United States and more than 230 million globally.¹ Infrapopliteal PAD, when symptomatic, most commonly manifests as chronic limb-threatening ischemia (CLTI). Surgical and endovascular revascularization carries a Class 1 recommendation in multispecialty consensus guidelines to alleviate rest pain, promote wound healing, reduce the risk for amputation, and improve quality of life in patients with infrapopliteal PAD and CLTI.^{2,3} Among available endovascular revascularization modalities, percutaneous transluminal angioplasty (PTA) with bail-out stenting is most widely used⁴ and recommended in U.S. and European guidelines. However, this approach may be limited by restenosis, requiring reintervention.^{5,6} Drug-eluting technologies, including drug-coated balloons (DCB) and drug-eluting stents (DES), may reduce the likelihood of restenosis,⁵ but the comparative efficacy of these modalities remains uncertain.^{5–8} We conducted a comprehensive network meta-analysis of published randomized controlled trials (RCTs), comparing the efficacy of endovascular therapies, including PTA, bare-metal stents (BMS), DCB angioplasty, and DES, for symptomatic infrapopliteal PAD.

METHODS

This study was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension statement for reporting of systematic reviews, incorporating network meta-analyses of health care interventions⁹ (Supplemental Table 1). The work was registered with International Prospective Register of Systematic Reviews (CRD42024583088).

DATABASES AND SEARCH STRATEGY. PubMed, the Web of Science, Cochrane Library, and Clinical-Trials.gov were searched from database inception through March 2025 for RCTs comparing endovascular therapies. Eligible studies compared PTA, BMS, DCBs, or DES, with bailout BMS or DES permitted after PTA or DCB angioplasty. Two authors (S.H.M. and N.Q.Q.) collaboratively developed a single search filter using relevant keywords and corresponding Medical Subject Headings terms (Supplemental

Table 2 for the search strategy). Non-English studies were eligible for inclusion if a full-text English translation was available.

ELIGIBILITY CRITERIA. Only RCTs that compared the efficacy of PTA, BMS, DCBs, or DES for treating infrapopliteal PAD were included. Studies reporting combined treatment of aortoiliac and/or femoropopliteal disease were excluded to maintain anatomical uniformity across studies. Those studies using carbon-plated or self-expanding Nitinol stents were excluded to adhere to a strict definition of BMS. Studies with sample sizes of fewer than 30 subjects were excluded to reduce small study effect variability. Inclusion and exclusion criteria for each study are presented in Supplemental Table 3.

STUDY SELECTION. Search results were exported to EndNote x9 (Clarivate Analytics). Titles and abstracts from search results were screened to identify relevant studies, and full texts of the selected studies were reviewed to assess eligibility for inclusion. The study selection process was completed independently by 2 authors (S.H.M. and N.Q.Q.), with conflicts resolved by a third reviewer (H.D.A.). The final list of studies to be included was reviewed and approved by 3 coauthors (S.H.M., N.Q.Q., and H.D.A.). The risk for bias within individual studies was evaluated using the Cochrane Collaboration’s risk-of-bias tool for randomized trials (RoB 2).

DATA COLLECTION. Data were independently extracted by 2 authors using Excel 2013 (Microsoft) and consolidated into a single .csv file. Extracted patient-, procedure-, and outcome-related variables appear in Tables 1 and 2. All analyses were conducted using available data, and no imputation was performed for missing data.

SUBJECTS. Patients with symptomatic infrapopliteal PAD undergoing endovascular revascularization (Rutherford classes 3–6) were included.

EXPOSURE. Primary planned revascularization modalities included PTA, BMS, DCB angioplasty, and DES. In the case of PTA or DCB angioplasty, bailout stenting was permissible in most studies to treat flow limiting dissection. Bioresorbable and coronary DES

ABBREVIATIONS AND ACRONYMS

ABI	= Ankle brachial index
BMS	= bare-metal stent(s)
CD-TLR	= clinically driven target lesion revascularization
CLTI	= chronic limb-threatening ischemia
DCB	= drug-coated balloon
DES	= drug-eluting stent(s)
LLL	= late lumen loss
MD	= mean difference
PAD	= peripheral artery disease
PTA	= percutaneous transluminal angioplasty
RCT	= randomized controlled trial
RVD	= reference vessel diameter
SUCRA	= surface under the cumulative ranking

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors’ institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

were included in the DES group. In the primary analysis, DCBs, BMS, and DES were compared with PTA. In secondary analyses, treatment modalities were compared with those other than PTA ([Supplemental Appendix](#)).

OUTCOMES. Angiographic outcomes included primary patency (as defined per study; individual study definitions appear in [Supplemental Table 4](#)), target lesion binary restenosis (defined as a $\geq 50\%$ reduction in luminal diameter in the treated segment, confirmed on ultrasound imaging or angiography), late lumen loss, and percentage in-lesion diameter stenosis. For analyses of angiographic outcomes, the number of lesions served as the denominator. Clinical outcomes included clinically driven target lesion revascularization (CD-TLR), a combination of clinical, angiographic and ultrasonographic criteria as defined in each study (individual study definitions appear in [Supplemental Table 5](#)), wound healing (among those in Rutherford class 5 and 6), target limb amputation, all-cause mortality, and improvement in Rutherford class.

STATISTICAL ANALYSIS. Random-effects models were used for primary, secondary, subgroup, and sensitivity analyses, with results reported as ORs for binary outcomes and mean differences (MDs) for continuous outcomes, each with its 95% CI. Four subgroup analyses were performed: 1) comparing bioresorbable DES with coronary DES; 2) comparing any drug-eluting technology (DCBs and DES) with non-drug-eluting technology (PTA and BMS); 3) comparing the antiproliferative agent used in patients with DES; and 4) stratifying by follow-up duration. Sensitivity analyses were conducted for all study endpoints by: 1) excluding drug-eluting bioresorbable scaffolds; and 2) excluding studies involving patients in Rutherford class 3. (Heterogeneity was assessed using Cochran's Q test and quantified using the I^2 statistic, categorized as low [$<25\%$], moderate [25% - 50%], or high [$>50\%$], with corresponding P values). Network geometry plots were created to display the number of studies with direct comparisons between treatments. The I^2 statistic and network plots are included with forest plots from the primary analysis. P values of <0.05 were considered to indicate statistical significance.

A separate indirect from direct evidence analysis was conducted for all outcomes using a back calculation approach under a random-effects model to evaluate consistency between direct (head-to-head) and indirect (using a common comparator) comparisons. Indirect estimates were generated only when a common comparator was present. Surface under the

cumulative ranking (SUCRA) p scores were calculated to rank treatments by efficacy. SUCRA p scores are relevant when differences in efficacy are significant but less so when differences are nonsignificant. A league table was generated to rank treatments on the basis of probabilities derived from rankograms, which reflect the likelihood of each treatment's being the best, second best, and so on. Funnel plots were used to assess publication bias, with significance evaluated using Egger's test. Results of separate indirect from direct evidence analysis, SUCRA ranking, league tables, and funnel plots are detailed sequentially for each outcome in the [Supplemental Appendix](#). The analysis was conducted using the netmeta package in R, with figures generated using ggplot2.

ETHICAL APPROVAL. This study is a network meta-analysis of previously published RCTs. It did not involve new data collection, direct interaction with human participants, or access to individual-level patient data. Therefore, Institutional Review Board approval and informed consent were not required. The meta-analysis was prepared using standard protocols and registered with the International Prospective Register of Systematic Reviews ([CRD42024583088](#)).

RESULTS

STUDY SELECTION. Twenty-one studies published since 2008 were included in the final analysis, as outlined in the Preferred Reporting Items for Systematic Reviews and Meta-Analyses flow diagram ([Figure 1](#)). Study characteristics appear in [Supplemental Table 6](#).

STUDY, PATIENT, AND PROCEDURAL CHARACTERISTICS.

A total of 2,984 patients were followed for a median of 12 months (range: 6-12 months) (13 trials for 12 months, 1 trial for 9 months, and 7 trials for 6 months). The mean age was 72.2 ± 9.5 years, and 71% ($n = 2,053$) were men. Hypertension (85.2% [$n = 2,425$]) and diabetes (72.8% [$n = 2,174$]) were the most prevalent comorbidities. Among the 16 trials ($n = 2,555$) that reported Rutherford classification, 6.8% of patients ($n = 173$) were in Rutherford class 3, 30.4% ($n = 778$) in Rutherford class 4, 59.4% ($n = 1,518$) in Rutherford class 5, and 3.4% ($n = 86$) in Rutherford class 6. The mean lesion length was 84.9 ± 61.3 mm (18 trials, $n = 2,667$), the mean baseline percentage diameter stenosis was $83.1\% \pm 17.6\%$ (15 trials, $n = 2,617$), and the proportion of chronic total occlusions was 37% (15 trials, $n = 826$). Additional patient and lesion characteristics appear in [Tables 1 and 2](#).

TABLE 1 Baseline Clinical Characteristics of Patients Included in the Network Meta-Analysis of Endovascular Treatments for Infrapopliteal Arterial Occlusive Disease

First Author	Trial Name	Year	Comparison Groups	Total Patients (Group 1/Group 2)	Primary Outcome Time Point, mo	Age, y	Men	DM
Varcoe a	LIFE-BTK	2024	DES vs PTA	261 (173/88)	12	72.6 ± 10.1	178 (68.2)	184 (70.5)
Overhagen ^a	SAVAL	2023	DES vs PTA	201 (130/71)	12	72.9 ± 9.9	150 (74.6)	127 (63.2)
Ahn ^a	SENS-BTK	2023	BMS vs PTA	119 (61/58)	12	67.7 ± 9.3	100 (84.0)	98 (82.4)
Liistro ^a	IN.PACT BTK	2022	DCB vs PTA	50 (23/27)	9	71.2 ± 8.5	39 (78.0)	43 (86.0)
Jia ^b	AcoArt II-BTK	2021	DCB vs PTA	120 (61/59)	12	70.7 ± 8.1	75 (62.5)	87 (72.5)
Patel ^b	SINGA-PACLI	2021	DCB vs PTA	138 (70/68)	12	62.5 ± 12.0	93 (67.3)	138 (100.0)
Liistro ^a	ACOART-BTK	2020	DCB Vs PTA	105 (52/53)	6	75.1 ± 8.7	80 (76.2)	101 (96.2)
Mustapha ^b	Lutonix BTK	2019	DCB vs PTA	442 (287/155)	6	72.9 ± 9.7	319 (72.2)	324 (73.3)
Haddad et al ^a	—	2017	DCB vs PTA	93 (45/48)	6	—	—	89 (95.6)
Spreen ^a	PADI	2016	DES vs PTA	137 (73/64)	12	73.5 ± 12.0	96 (70.1)	87 (63.5)
Zeller ^b	BIOLUX P-II	2015	DCB vs PTA	72 (36/36)	12	71.3 ± 9.6	57 (79.1)	48 (66.6)
Siablis ^b	IDEAS	2014	DES vs DCB	50 (25/25)	6	71.5 ± 9.6	38 (76.0)	35 (70.0)
Zeller ^b	IN.PACT DEEP	2014	DCB vs PTA	358 (239/119)	12	72.8 ± 9.3	266 (74.3)	263 (73.4)
Liistro ^a	DEBATE-BTK	2013	DCB vs PTA	132 (65/67)	12	74.5 ± 9.5	106 (80.3)	132 (100.0)
Bosiers ^a	DESTINY	2012	DES vs BMS	140 (74/66)	12	76.0 ± 8.2	89 (63.5)	77 (55.0)
Scheinert ^b	ACHILLES	2012	DES vs PTA	200 (99/101)	12	73.4 ± 8.8	143 (71.5)	129 (64.5)
Rastan ^a	YUKON-BTK	2011	DES vs BMS	161 (82/79)	12	72.9 ± 9.0	108 (67.1)	87 (54.0)
Brodmann et al ^a	—	2011	PTA vs BMS	54 (33/21)	12	72.6 ± 1.4	25 (46.3)	40 (74.1)
Randon et al ^a	—	2010	PTA vs BMS	38 (22/16)	6	72 ± 9.8	20 (52.6)	22 (57.8)
Tepe ^a	BELOW	2010	DES vs BMS vs PTA	63 (14/16/33)	6	72.9	42 (66.7)	43 (68.5)
Falkowshki ^b	—	2008	DES vs BMS	50 (25/25)	6	69.0 ± 4.0	29 (58.0)	20 (40.0)
Weighted averages	—	—	PTA vs BMS vs DCB vs DES	2,984 (1,102/284/903/695)	—	72.2 ± 9.5	2,053 (71.0)	2,174 (72.8)

Values are mean ± SD or n (%). Dashes represent missing data. ^aStudy enrolled patients in Rutherford classes >4. ^bStudy enrolled patients in Rutherford classes >3. ^cMean Rutherford classification was 4.5 (4-5) and 4.5 (3-5) in the DES and PTA arms, respectively. ^dMean Rutherford classification was 4.1 ± 0.9 and 4.0 ± 0.9 in the stent and PTA arms, respectively.

BMS = bare-metal stent(s); CAD = coronary artery disease; DCB = drug-coated balloon; DES = drug-eluting stent(s); DM = diabetes mellitus; HLD = hyperlipidemia; HTN = hypertension; PTA = percutaneous transluminal angioplasty.

Continued on the next page

Across the 21 studies, 17 treated patients with PTA, 7 with BMS, 10 with DCBs, and 9 with DES. All 10 DCB studies used paclitaxel-coated balloons; among studies using DES, paclitaxel was used in 2, sirolimus in 5, everolimus in 3, and zotarolimus in 1.

ANGIOGRAPHIC OUTCOMES. Primary patency (20 studies, n = 2,295). The primary patency rates at a median of 12 months were 48.5% (n = 426) for PTA, 55.7% (n = 83) for BMS, 68.2% (n = 461) for DCBs, and 72.8% (n = 431) for DES. DCBs (OR: 2.66; 95% CI: 1.52-4.64; *P* < 0.001) and DES (OR: 2.41; 95% CI: 1.23-4.73; *P* = 0.010) had significantly higher primary patency compared with PTA, whereas BMS was not significantly different from PTA (OR: 0.58; 95% CI: 0.24-1.42; *P* = 0.23) (Figure 2A, Supplemental Figures 1 to 5).

Target lesion binary (>50%) restenosis (16 studies, n = 1,672). Target lesion binary restenosis rates were 56.5% (n = 354) for PTA, 45.7% (n = 86) for BMS, 43.5% (n = 166) for DCBs, and 20.6% (n = 110) for DES. Compared with PTA, DCBs (OR: 0.40;

95% CI: 0.21-0.77; *P* = 0.006) and DES (OR: 0.28; 95% CI: 0.13-0.62; *P* = 0.002) had significantly lower restenosis rates, whereas rates for BMS were not significantly different from those for PTA (OR: 1.39; 95% CI: 0.49-3.93; *P* = 0.54) (Figure 2B, Supplemental Figures 6 to 10).

Late lumen loss (8 studies, n = 742). The average late lumen loss was 0.89 ± 0.75 mm for PTA, 1.52 ± 0.91 mm for BMS, 0.61 ± 0.68 mm for DCBs, and 0.79 ± 0.72 mm for DES. Compared with PTA, late lumen loss was significantly higher for primary BMS (MD: 0.87; 95% CI: 0.18-1.58; *P* = 0.01), significantly lower with DCBs (MD: -0.34; 95% CI: -0.63 to -0.05; *P* = 0.02), and similar with DES (MD: 0.03; 95% CI: -0.50 to 0.45; *P* = 0.91) (Figure 2C, Supplemental Figures 11 to 15).

Percentage in-lesion diameter stenosis (11 studies, n = 1,246). The percentage in-lesion diameter stenosis was 44.0% ± 35.7% for PTA, 41.8% ± 25.6% for BMS, 33.7% ± 25.3% for DCBs, and 29.0% ± 24.4% for DES. Compared with PTA, the rate of in-lesion stenosis was not different with BMS (MD:

TABLE 1 Continued							
HTN	HLD	Smoker	CAD	Rutherford Class			
				3	4	5	6
243 (93.1)	212 (81.2)	138 (52.8)	40 (15.3)	—	135 (51.7)	126 (48.3)	—
174 (86.6)	153 (76.1)	144 (71.6)	—	—	107 (53.2)	93 (46.3)	—
77 (64.7)	25 (21.0)	42 (35.3)	31 (26.1)	—	42 (35.2)	76 (63.8)	1 (0.8)
40 (80.0)	36 (72.0)	7 (14.0)	19 (38.0)	—	6 (12.0)	44 (88.0)	—
94 (78.3)	41 (34.2)	32 (26.7)	42 (35.0)	1 (0.8)	51 (42.5)	52 (43.3)	16 (13.3)
115 (83.3)	117 (84.8)	49 (35.5)	86 (62.3)	5 (3.6)	100 (72.5)	33 (23.9)	—
87 (82.3)	60 (57.1)	52 (49.6)	36 (34.3)	—	8 (7.6)	92 (87.6)	9 (8.6)
431 (97.5)	356 (80.5)	271 (61.3)	259 (58.6)	44 (9.9)	152 (34.3)	266 (60.2)	—
79 (84.9)	85 (91.4)	—	—	—	—	—	—
—	—	63 (45.9)	52 (37.9)	—	26 (18.9)	94 (68.6)	28 (20.4)
62 (86.1)	49 (68.1)	40 (55.5)	30 (41.7)	12 (16.7)	4 (5.5)	52 (72.2)	—
25 (50.0)	23 (46.0)	15 (30.0)	14 (28.0)	— ^c	—	—	—
320 (89.3)	255 (71.2)	52 (14.5)	—	1 (0.3)	55 (15.4)	293 (81.8)	9 (2.5)
98 (74.2)	39 (29.6)	20 (15.1)	22 (16.7)	—	5 (3.7)	115 (87.1)	23 (17.4)
96 (68.5)	53 (37.8)	45 (32.1)	—	—	63 (45.0)	77 (55.0)	—
181 (90.5)	146 (73.0)	65 (32.5)	130 (65.0)	— ^d	—	—	—
145 (90.0)	123 (76.4)	46 (28.6)	—	76 (47.2)	9 (5.5)	66 (40.9)	—
46 (85.2)	20 (37)	15 (27.8)	47 (87)	—	—	—	—
38 (100.0)	16 (42.1)	5 (13.2)	30 (78.9)	—	5 (13.2)	33 (84.6)	—
43 (68.5)	22 (34.9)	4 (6.3)	—	—	—	—	—
31 (62.0)	18 (36.0)	22 (44.0)	21 (42.0)	34 (68.0)	10 (20.0)	6 (12.0)	—
2,425 (85.2)	1,849 (64.9)	1,127 (39.0)	860 (37.0)	173 (6.8)	778 (30.4)	1,518 (59.4)	86 (3.4)

12.31; 95% CI: −8.64 to 33.25; $P = 0.25$), significantly lower with DCB (MD: −10.04; 95% CI: −19.20 to −0.88; $P = 0.03$), and numerically but not significantly lower with DES (MD: −13.29; 95% CI: −28.78 to 2.20; $P = 0.09$) (Figure 2D, Supplemental Figures 16 to 20).

CLINICAL OUTCOMES. CD TLR (20 studies, n = 2,739). CD-TLR rates were 21.1% (n = 215) for PTA, 27.2% (n = 71) for BMS, 14.7% (n = 120) for DCBs, and 8.8% (n = 57) for DES. Compared with PTA, rates of CD-TLR were numerically but not significantly greater with BMS (OR: 1.95; 95% CI: 0.90-4.21; $P = 0.08$), significantly lower with DCBs (OR: 0.47; 95% CI: 0.29-0.76; $P = 0.002$), and numerically but not significantly lower with DES (OR: 0.57; 95% CI: 0.29-1.10; $P = 0.09$) (Figure 3A, Supplemental Figures 21 to 25).

Wound healing (15 studies, n = 1,650). Complete wound healing was observed in 64.7% of patients (n = 339) for PTA, 55.7% (n = 54) for BMS, 67.7% (n = 442) for DCBs, and 52.9% (n = 138) for DES. Compared with PTA, the proportion of patients achieving complete wound healing was significantly lower with BMS (OR: 0.41; 95% CI: 0.19-0.85; $P = 0.017$) and similar with DCBs (OR: 1.23; 95% CI: 0.79-1.90; $P = 0.36$)

and DES (OR: 0.92; 95% CI: 0.53-1.60; $P = 0.77$) (Figure 3B, Supplemental Figures 26 to 30).

Target limb amputation (19 studies, n = 2,754). Target limb amputation rates were 5.7% (n = 58) for PTA, 3.7% (n = 8) for BMS, 4.9% (n = 44) for DCBs, and 5.0% (n = 31) for DES and did not differ significantly among the 4 groups. Compared with PTA, there were no significant differences in the likelihood of target limb major amputation for BMS (OR: 1.29; 95% CI: 0.43-3.90; $P = 0.65$), DCBs (OR: 1.29; 95% CI: 0.74-2.24; $P = 0.37$), or DES (OR: 0.73; 95% CI: 0.42-1.26; $P = 0.25$) (Figure 3C, Supplemental Figures 31 to 35).

All-cause mortality (19 studies, n = 2,768). The average 1-year mortality across all studies was 11.3%, with rates of 11.3% (n = 114) for PTA, 21.1% (n = 41) for BMS, 8.6% (n = 77) for DCBs, and 12.0% (n = 80) for DES. Compared with PTA, there were no significant differences in all-cause mortality for BMS (OR: 0.94; 95% CI: 0.53-1.67; $P = 0.83$), DCBs (OR: 1.11; 95% CI: 0.76-1.62; $P = 0.60$), or DES (OR: 0.96; 95% CI: 0.64-1.45; $P = 0.84$) (Figure 3D, Supplemental Figures 36 to 40).

Mean improvement in Rutherford classification (11 studies, n = 1,289). The mean improvement in Rutherford classification was 3.07 ± 1.19 for PTA,

TABLE 2 Baseline Angiographic Characteristics of Treated Lesions Included in the Network Meta-Analysis of Endovascular Treatments for Infrapopliteal Arterial Occlusive Disease

First Author (Trial Name)	Total Lesions	ABI	Lost to FUP	Lesion Length, mm	Diameter Stenosis, %	RVD, mm	Vessel Calcification ^a	Total Occlusion
Varcoe (LIFE-BTK) ^b	271	0.9 ± 0.3	9	46.9 ± 30.5	73.0 ± 19.6	2.9 ± 0.8	82 (30.3)	118 (43.5)
Overhagen (SAVAL) ^b	218	0.9 ± 0.3	—	68.4 ± 42.2	77.6 ± 17.1	3.1 ± 0.7	64 (29.4)	65 (29.8)
Ahn (SENS-BTK) ^b	119	0.9 ± 0.3	0	—	—	—	—	—
Liistro (IN.PACT BTK) ^b	55	—	0	216.8 ± 82.1	96.9 ± 7.8	2.8 ± 0.5	6 (10.9)	55 (100.0)
Jia (AcoArt II-BTK) ^c	131	0.5 ± 0.3	5	174.9 ± 83.3	96.0 ± 13.2	2.5 ± 0.4	12 (9.2)	102 (77.8)
Patel (SINGA-PACLI) ^c	136	0.8 ± 0.3	21	86.1 ± 72.9	78.0 ± 19.9	2.2 ± 0.6	88 (64.7)	44 (32.4)
Liistro (ACOART-BTK) ^b	128	0.4 ± 0.2	0	177.5 ± 111	91.2 ± 14.4	2.71 ± 0.3	82 (64.1)	86 (67.2)
Mustapha (Lutonix BTK Trial) ^c	680	—	33	103.3 ± 89.0	85.8 ± 15.2	2.4 ± 0.5	63 (9.3)	222 (32.6)
Haddad ^b	93	—	0	—	—	—	—	—
Spreen (PADI) ^b	212	—	—	22.1 ± 20.6	83.2 ± 16.0	2.9 ± 0.6	—	—
Zeller (BIOLUX P-II) ^c	104	—	5	114.1 ± 87.5	74.3 ± 24.3	2.2 ± 0.6	19 (18.2)	43 (41.3)
Siablis (IDEAS) ^c	55	—	4	137.5 ± 51.6	86.1 ± 9.5	—	26 (45.5)	10 (18.2)
Zeller (IN.PACT DEEP) ^c	531	0.8 ± 0.4	81	115.1 ± 92.8	84.8 ± 16.5	2.4 ± 0.7	67 (12.7)	106 (19.9)
Liistro (DEBATE-BTK) ^b	158	0.3 ± 0.3	7	130.0 ± 81.0	97.2 ± 7.8	2.9 ± 0.3	42 (26.5)	126 (79.7)
Bosiers (DESTINY) ^b	154	—	7 (5)	17.4 ± 10.1	78.5 ± 14.5	3.0 ± 0.8	116 (75.3)	25 (16.2)
Scheinert (ACHILLES) ^c	228	—	16	26.9 ± 21.1	71.4 ± 19.2	2.6 ± 0.5	35 (15.4)	178 (78.1)
Rastan (YUKON-BTK) ^b	161	0.5 ± 0.2	8	30.0 ± 8.5	88.0 ± 9.0	3.0 ± 0.4	—	36 (22.4)
Brodmann ^b	54	—	0	58.8 ± 86.6	—	—	—	16 (30)
Randon ^b	56	—	0	—	—	—	—	—
Tepe (BELOW) ^b	63	—	7	33.3 ± 18.5	—	2.9 ± 0.2	—	14 (22.2)
Falkowshki ^c	50	—	0	17.8 ± 3.4	—	2.7 ± 1.7	—	—
Weighted averages	3,749	0.7 ± 0.3	214 (9.0)	84.9 ± 61.3	83.1 ± 17.6	2.9 ± 0.3	702 (24.6)	1,246 (39.8)

Values are mean ± SD or N (%). ^aVessel calcification was defined as the proportion of patients with moderate to severe calcification. ^bEnrolled patients in Rutherford class >4. ^cEnrolled patients in Rutherford class >3.

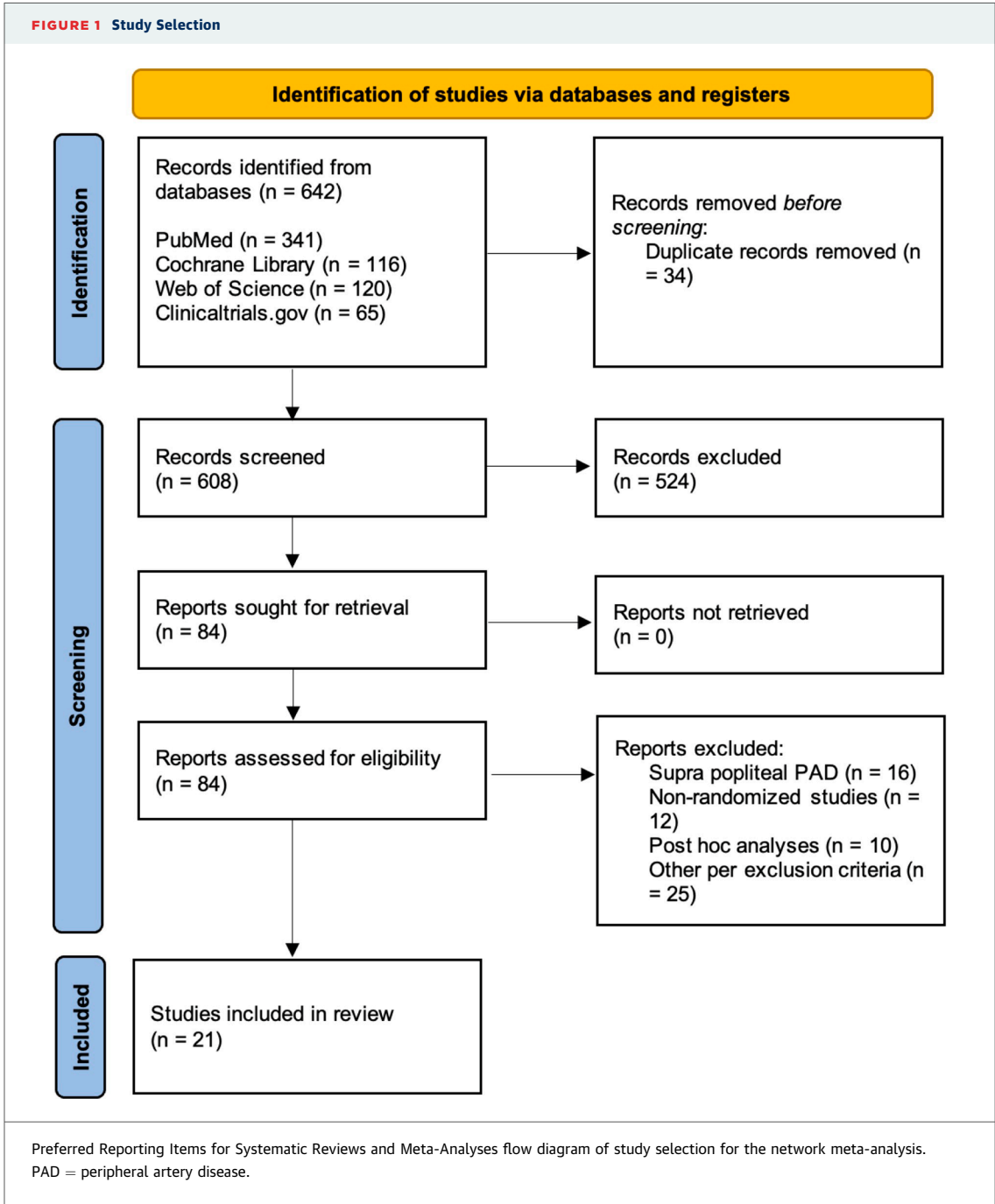
ABI = ankle-brachial index; FUP = follow-up; RVD = reference vessel diameter.

2.05 ± 1.36 for BMS, 2.92 ± 1.66 for DCBs, and 2.98 ± 0.67 for DES and did not differ significantly among the 4 groups. Compared with PTA, there were no significant differences with BMS (MD: 0.27; 95% CI: -1.24 to 0.72; $P = 0.59$), DCB (MD: -0.05; 95% CI: -0.53 to 0.43; $P = 0.84$), or DES (MD: 0.01; 95% CI: -0.59 to 10.57; $P = 0.97$) (Figure 3E, Supplemental Figures 41 to 45).

SUBGROUP ANALYSIS. Bioresorbable DES vs Coronary DES (n = 173 vs n = 522 patients). There was no difference in primary patency (OR: 1.08; 95% CI: 0.17-6.91; $P = 0.93$; $I^2 = 75.1\%$) or target lesion binary restenosis (OR: 1.53; 95% CI: 0.20-11.86; $P = 0.69$; $I^2 = 76.6\%$) between bioresorbable and coronary DES (Supplemental Figure 46). Similarly, no differences were noted in CD-TLR rates (OR: 0.61; 95% CI: 0.12-3.18; $P = 0.20$; $I^2 = 56.0\%$), complete wound healing (OR: 0.61; 95% CI: 0.17-2.17; $P = 0.45$; $I^2 = 42.9\%$), target limb amputation (OR: 6.93; 95% CI: 0.35-137.31; $P = 0.20$; $I^2 = 53.6\%$), or mortality (OR: 1.34; 95% CI: 0.42-4.29; $P = 0.62$; $I^2 = 51.1\%$) (Supplemental Figure 47).

DRUG-ELUTING VS NON-DRUG-ELUTING (n = 1,598 vs n = 1,386 PATIENTS). Drug-eluting technology (DCBs and DES) was associated with significantly higher rates of primary patency (OR: 1.44; 95% CI: 1.23-1.67; $P < 0.0001$; $I^2 = 76.4\%$) and a lower incidence of target lesion binary restenosis (OR: 0.57; 95% CI: 0.45-0.72; $P < 0.0001$; $I^2 = 72.4\%$), late lumen loss (MD: -0.43; 95% CI: -0.73 to -0.13; $P = 0.005$; $I^2 = 87.2\%$), and percentage diameter stenosis (MD: -13.4; 95% CI: -21.77 to -5.00; $P = 0.01$; $I^2 = 93.3\%$) compared with non-drug-eluting technology (PTA and BMS) (Supplemental Figure 48).

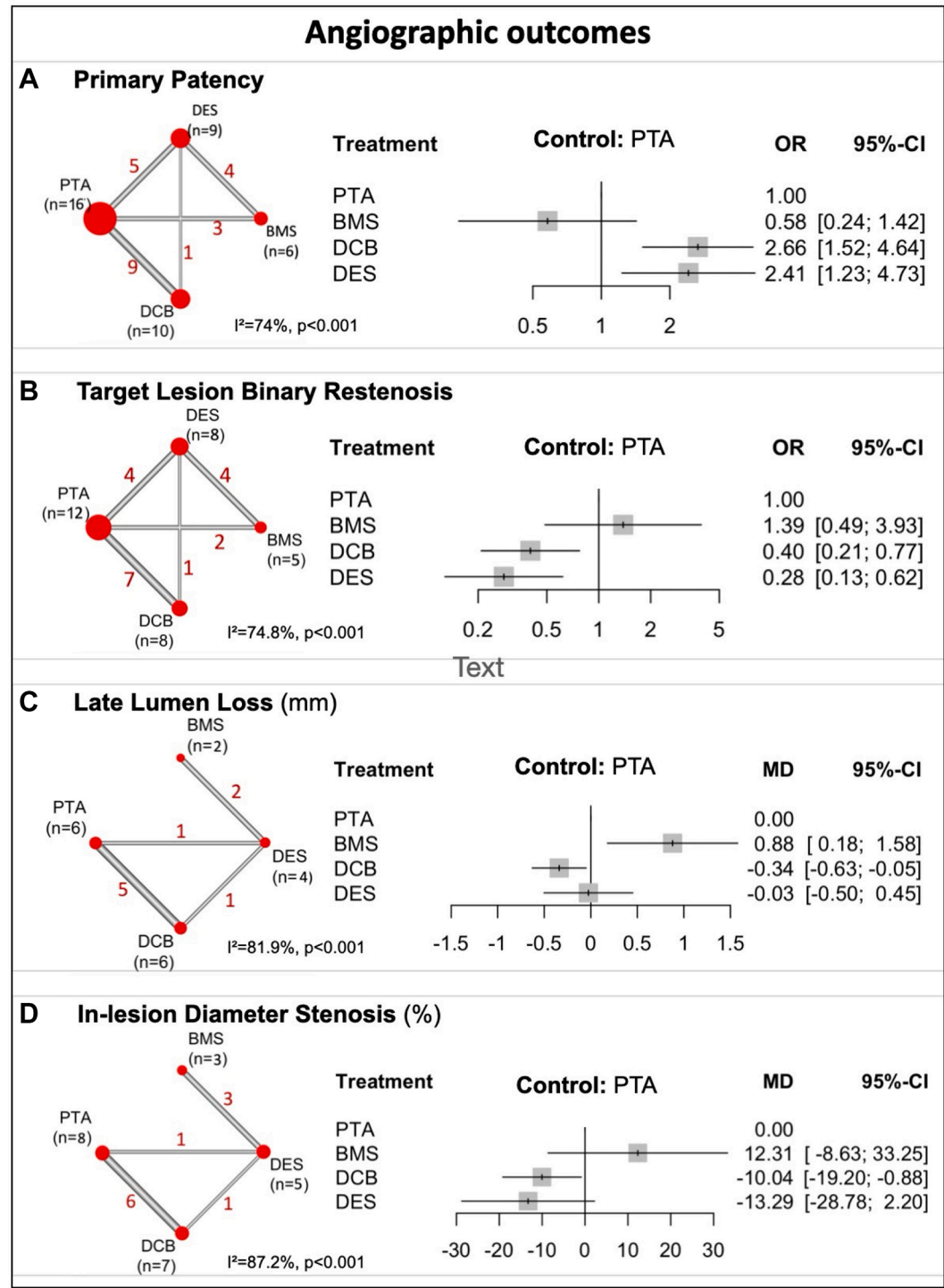
Drug-eluting technology had lower rates of CD-TLR (OR: 0.54; 95% CI: 0.40-0.73; $P < 0.001$; $I^2 = 53.2\%$) compared with non-drug-eluting technology. There was no significant difference in wound healing (OR: 1.07; 95% CI: 0.96-1.18; $P = 0.23$; $I^2 = 43.1\%$), target limb amputation (OR: 0.95; 95% CI: 0.67-1.34; $P = 0.75$; $I^2 = 0\%$), all-cause mortality (OR: 1.04; 95% CI: 0.82-1.30; $P = 0.76$; $I^2 = 0\%$), or mean improvement in Rutherford classification (MD: 0.03; 95% CI: -0.32 to 0.37; $P = 0.88$; $I^2 = 97.2\%$) between the 2 subgroups (Supplemental Figure 49).



Comparison of antiproliferative agent used in DES: paclitaxel (n = 203) vs sirolimus (n = 233) vs everolimus (n = 345 patients). No differences were observed among sirolimus-, everolimus-, and paclitaxel-coated stents in rates of primary patency, target lesion binary restenosis, CD-TLR, target limb amputation, all-cause mortality, or improvement in Rutherford classification ([Supplemental Figures 50 and 51](#)). Studies using DES typically required longer

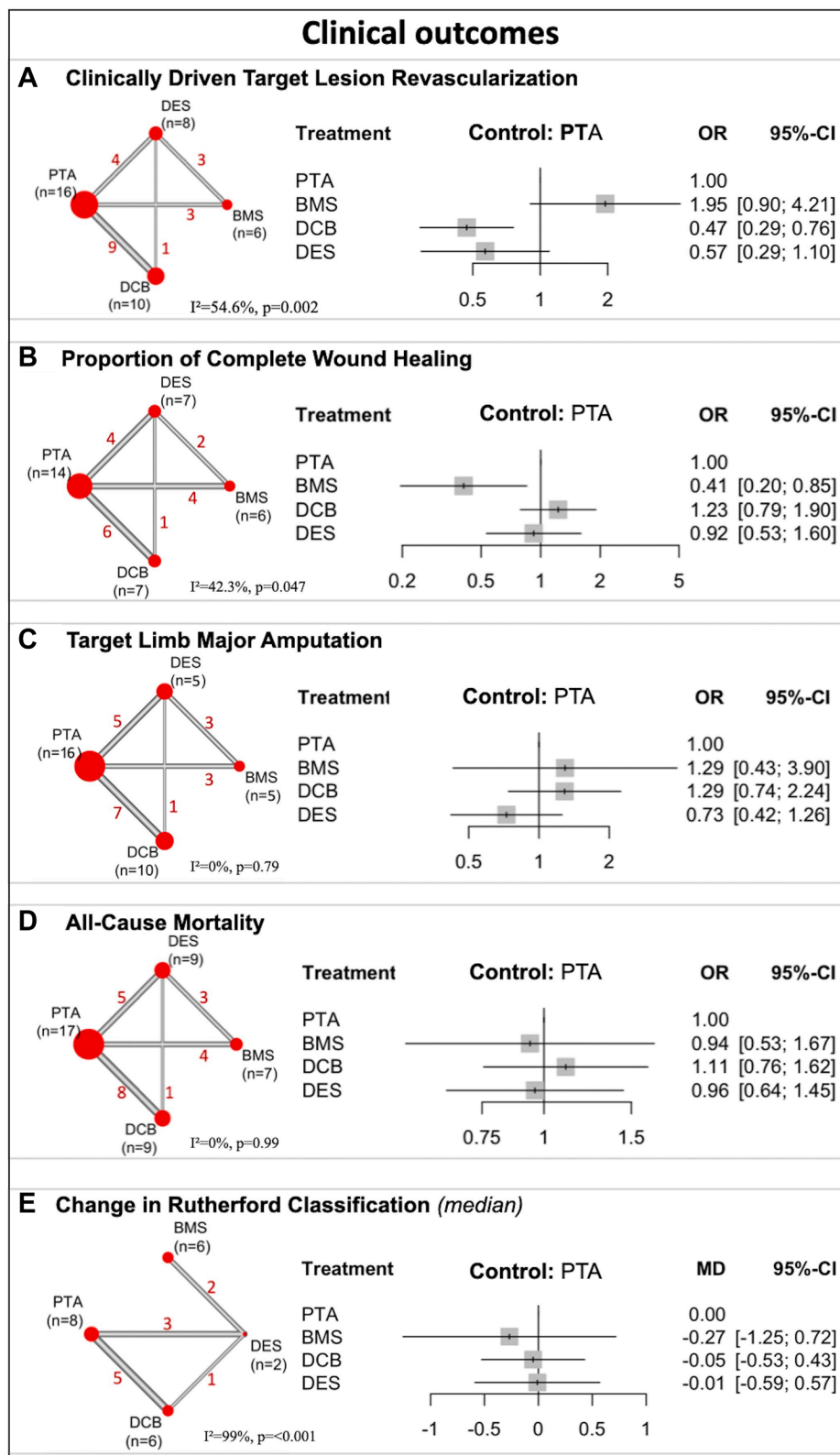
durations of dual antiplatelet therapy (median 6-12 months), while those using DCB or PTA reported more variable and shorter durations (median 1-6 months), often guided by institutional protocols ([Supplemental Table 6](#)).
Stratification by follow-up duration: 12 months (n = 13 trials) and <12 months (n = 8 trials). At 12-month follow-up, compared with PTA, DCBs had superior primary patency (OR: 1.47; 95% CI: 1.06-

FIGURE 2 Angiographic Outcomes



Forest plots and network geometry depicting effect sizes with 95% CIs for angiographic outcomes at a median follow-up of 12 months, with percutaneous transluminal angioplasty (PTA) as the reference. BMS = bare-metal stent(s); DCB = drug-coated balloon; DES = drug-eluting stent(s).

FIGURE 3 Clinical Outcomes



Forest plots and network geometry depicting effect sizes with 95% CIs for clinical outcomes at a median follow-up of 12 months, with PTA as the reference. Abbreviations as in [Figure 2](#).

2.04; $P = 0.02$; $I^2 = 78.8\%$), and DES had significantly lower target lesion binary restenosis (OR: 0.55; 95% CI: 0.34–0.87; $P = 0.003$; $I^2 = 0\%$) and lower percentage in-lesion diameter stenosis (MD: -12.80 ; 95% CI: -19.76 to -5.84 ; $P < 0.001$; $I^2 = 0\%$). At a follow-up duration <12 months, DCB and DES had superior primary patency (OR: 1.62 [95% CI: 1.13–2.33; $P = 0.009$; $I^2 = 70.2\%$] and OR: 1.99 [95% CI: 1.18–3.35; $P = 0.01$; $I^2 = 70.2\%$], respectively) and lower target lesion binary restenosis (OR: 0.54 [95% CI: 0.38–0.76]; $P < 0.001$; $I^2 = 47.9\%$] and OR: 0.23 [95% CI: 0.12–0.47]; $P < 0.001$; $I^2 = 47.9\%$], respectively) than PTA. DCBs also had lower late lumen loss (MD: -0.66 ; 95% CI: -1.02 to -0.30 ; $P < 0.001$; $I^2 = 54.5\%$) (Supplemental Figure 52).

At 12-month follow-up, there were no differences in clinical outcomes among the 4 treatment groups. At a follow-up duration <12 months, DCB had lower rates of CD-TLR than PTA (OR: 0.44; 95% CI: 0.23–0.85; $P = 0.01$; $I^2 = 63.1\%$). BMS (MD: -1.23 ; 95% CI: -1.57 to -0.89 ; $P < 0.001$; $I^2 = 0\%$), DCBs (MD: -0.47 ; 95% CI: -0.67 to -0.27 ; $P < 0.001$; $I^2 = 0\%$), and DES (MD: -0.47 ; 95% CI: -0.78 to -0.16 ; $P < 0.001$; $I^2 = 0\%$) fared worse than PTA with respect to change in Rutherford class, and outcome reported in only 4 studies ($n = 560$) (Supplemental Figure 53).

SENSITIVITY ANALYSES. Exclusion of bioresorbable DES. Primary patency, target lesion binary restenosis, CD-TLR, wound healing, target limb major amputation, and all-cause mortality were unchanged (Supplemental Figures 54 and 55).

Exclusion of Rutherford class 3. Primary patency, target lesion binary restenosis, late lumen loss, percentage diameter stenosis, CD-TLR, target limb amputation, all-cause mortality, and improvement in Rutherford classification were unchanged. However, BMS had a significantly lower proportion of complete wound healing compared with PTA, while DCBs and DES showed no significant difference (Supplemental Figures 56 and 57).

BIAS ASSESSMENT WITHIN STUDIES. Results of the RoB 2 tool are presented in Supplemental Figures 58 and 59. Bias in the selection of reported result was the most common. Despite these limitations, inconsistency between direct and indirect comparisons was insignificant ($P > 0.05$ for all comparisons).

DISCUSSION

In this network meta-analysis of patients with symptomatic infrapopliteal PAD, we compared angiographic and clinical outcomes at a median of

12 months following randomization to PTA, BMS, DCBs, or DES. We observed that when compared with PTA, DCBs and DES had significantly higher primary patency and lower target lesion binary restenosis; likewise, DCBs had less late lumen loss, percentage in-lesion diameter stenosis, and CD-TLR. BMS performed either no differently or worse than PTA on both angiographic and clinical endpoints. Despite the favorable differences in outcome effected by DCB and DES, important clinical outcomes, including change in Rutherford classification, wound healing, target limb amputation, and all-cause mortality, were not different across treatment modalities (Central Illustration).

The most efficacious revascularization modality for treating infrapopliteal PAD remains unknown. Several head-to-head trials have compared PTA and DCB with provisional bail-out stenting vs planned BMS or DES; these have reported inconsistent results, and the findings of meta-analyses have also been mixed.^{5–8,10} The present meta-analysis is the most comprehensive performed to date, incorporating the recently published SAVAL and LIFE-BTK trials. We report the broadest range of angiographic and clinical endpoints in the largest pooled cohort of patients with infrapopliteal PAD. Additionally, we compared drug-eluting and non-drug-eluting technologies, investigated the role of bioresorbable vascular scaffolds, and examined revascularization efficacy in the setting of CLTI, areas not addressed in prior network meta-analyses. A summary of prior literature is provided in Supplemental Table 7.

PTA remains the oldest, most widely used revascularization modality for infrapopliteal PAD.⁴ U.S. and European guidelines advocate that PTA should serve as first-line therapy in infrapopliteal PAD because lesions are frequently diffuse.^{11,12} However, PTA is limited by high restenosis rates, often requiring reintervention. BMS were introduced to scaffold vessels after PTA, but evidence to support their long-term benefit is lacking. In fact, when stents are used in contemporary practice, a minority use BMS.¹³ Drug-eluting technology was introduced to address restenosis through delivery of an anti-proliferative agent, which has resulted in improved patency and lower reintervention rates. We believe our primary findings support DCB and/or DES, rather than PTA, as first-line options for patients with infrapopliteal PAD given their resultant improvements in angiographic outcomes and CD-TLR.

Nevertheless, it is notable that the meaningful differences in angiographic outcomes and CD-TLR seen with drug-eluting technologies did not translate into better wound healing or lower amputation

CENTRAL ILLUSTRATION Study Overview and Findings

Comparative Efficacy of Endovascular Revascularization Modalities for Infrapopliteal Peripheral Artery Disease: A Network Meta-Analysis

Overview of Included Studies

2,984 Patients; 21 RCTs
Mean age: 72.2 ± 9.5 years; 71% Male

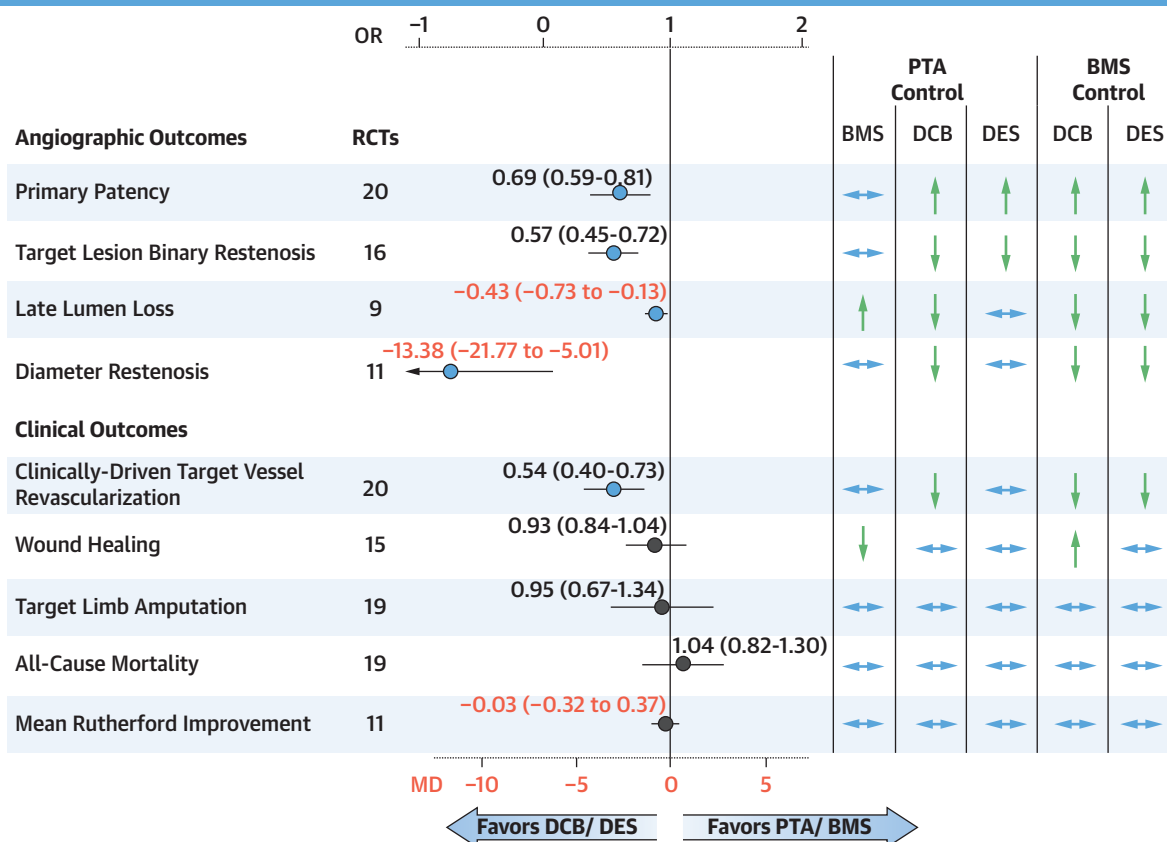
PTA
(16 RCTs, n = 1,102)

BMS
(7 RCTs, n = 284)

DCB
(10 RCTs, n = 903)

DES
(9 RCTs, n = 695)

Outcomes



- Trials comparing DCB or DES vs PTA found greater patency and less restenosis with drug-containing devices.
- For studies assessing DCB vs PTA, lower rates of LLL, diameter stenosis, and TLR were found with DCB.

● $P \leq 0.05$ ● $P > 0.05$ ↔ No Difference ↑ Higher ↓ Lower

Mufarrih SH, et al. JACC Cardiovasc Interv. 2026;19(13):1780-1792.

This illustration summarizes a network meta-analysis of 21 randomized controlled trials (RCTs) including 2,984 patients comparing percutaneous transluminal angioplasty (PTA), bare-metal stents (BMS), drug-coated balloons (DCBs), and drug-eluting stents (DES) for infrapopliteal peripheral artery disease. The figure displays pooled effect estimates for angiographic and clinical outcomes, with ORs or mean differences (MDs) presented relative to comparator strategies and directional comparisons shown against PTA and BMS. CD-TLR = clinically driven target lesion revascularization; LLL = late lumen loss.

rates during follow-up, especially as the primary goal of infrapopliteal revascularization among patients with CLTI is limb preservation. Prior studies have demonstrated that durable patency is an important contributor to wound healing and limb preservation, and that loss of patency may promote wound progression and limb loss.^{14,15} There are multiple potential explanations for the lack of concordance between angiographic and clinical outcomes in our study. First, pedal perfusion pressures required to heal tissue are greater than those to maintain tissue integrity; a transient rise in pedal pressure following any of the studied revascularization modalities may have been sufficient to facilitate wound closure, even if differential rates of restenosis or occlusion were observed later. Second, it was not possible to discern the effectiveness with which in-line blood flow was restored to wound beds nor whether adequate perfusion was restored to the foot, or the entire limb. Third, both wound healing and need for amputation, are influenced by systemic comorbidities, nutritional status, treatment of superimposed infection, and local interventions such as pressure offloading and wound care; the effectiveness of these and other postrandomization interventions was not routinely characterized in the included studies.¹⁶

STUDY LIMITATIONS. First, the patient populations studied, their lesion characteristics (eg, length, presence of calcification), and the stent types used differed across studies. For example, DCBs were often used in trials enrolling patients with longer, more diffuse lesions, whereas DES studies included patient with more focal disease with greater heterogeneity in antiproliferative agents. These differences may have contributed to variability in angiographic outcomes. To address this, we used a random-effects model and observed consistent results between direct and indirect comparisons despite high heterogeneity; nevertheless, indirect comparisons across devices should always be interpreted with caution.

Second, most included studies were small, single-center RCTs, such that variations in primary endpoint definition, reporting, and assessment may have influenced the overall results. Although our assessment revealed a positive publication bias specifically for in-lesion diameter stenosis, suggesting that smaller studies may have reported exaggerated effect sizes because of variability in outcome assessment, other endpoints were not similarly

affected. Despite being the largest analysis of its kind, this study may have been underpowered to detect differences in clinical outcomes. Third, there was a notable difference in pooled sample sizes of the 4 treatment groups, which may influence results. Fourth, most trials reported outcomes only to 12 months, potentially missing differences in long-term durability. Fifth, although wound-healing outcomes were consistently reported, only a small number of studies recorded rest-pain resolution, limiting meaningful comparisons in patients with Rutherford class 4 disease. Finally, translation of study findings to practice may be more challenging for U.S. operators given that smaller diameter DCBs have not yet received U.S. regulatory approval. Future research should include large, multicenter trials with extended follow-up that validate and/or extend our findings; investigate outcomes of DCBs containing various antiproliferative agents; examine bioresorbable, longer, and more flexible DES; and assess their long-term efficacy with respect to limb preservation and quality of life.

CONCLUSIONS

In the most comprehensive meta-analysis of patients with infrapopliteal PAD conducted to date, we found that compared with PTA, DCBs and DES demonstrated superior primary patency and lower target lesion binary restenosis, while DCBs reduced late lumen loss, in-lesion diameter stenosis, and CD-TLR. No advantage was apparent with BMS.

FUNDING SUPPORT AND AUTHOR DISCLOSURES

Dr Aronow is a consultant for Amplitude Vascular Systems, Medtronic, Novartis, Philips, RapidAI, Recor, and Silk Road Medical. Dr Secemsky is a consultant for Abbott Laboratories, Asahi, Bayer Healthcare Pharmaceuticals, Boston Scientific, Conavi, Cordis, Endovascular Engineering, Evident Engineering, Gore, Inari, InfraRedx, Janssen Pharmaceuticals, Medtronic, Penumbra, Philips, R3 Vascular, Rampart, RapidAI, Regeneron, Shockwave, Siemens, Teleflex, Terumo, Thrombolex, Venture Medical Group, and Zoll. Dr Secemsky receives research funding from Abbott Laboratories, AstraZeneca Pharmaceuticals, BD Bard, Boston Scientific, Cook, Laminar Medical, Medtronic, and Philips; and is on Speakers Bureau for BD Bard and Cook. All other authors have reported that they have no relationships relevant to the contents of this paper to disclose.

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PERSPECTIVES

WHAT IS KNOWN? Prior randomized trials have reported heterogeneity in angiographic and clinical outcomes across endovascular treatment modalities for symptomatic infrapopliteal PAD.

WHAT IS NEW? In this network meta-analysis, DCBs and DES were associated with superior primary patency and lower rates of binary restenosis, and DCBs were associated with less late lumen loss, lower percentage in-lesion diameter stenosis, and less CD TLR, compared with

PTA, whereas BMS offered no clinical or angiographic advantage. Despite these benefits, improvements in vessel patency were not associated with improvement in other clinical outcomes.

WHAT IS NEXT? Future studies should be powered to detect differences in a broader array of clinical outcomes, use perfusion-based definitions of procedural success, and mandate standardized approaches to postprocedural wound care.

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KEY WORDS angioplasty, below knee, drug-eluting stent, drug-coated balloon, infrapopliteal, peripheral artery disease, revascularization, stent

APPENDIX For supplemental methods, tables, results, figures, discussion, and references, please see the online version of this paper.